Research Paper Review / Literature Survey

Problem Statement: Smart High-Speed Rail Passenger Experience Platform

CO5 Assessment Tool 1 — Literature Survey (CSA10 / CSA1063 – Software Engineering)

Course Code:	CSA1007	**Course Title:**	Software Engineering
CO Assessed:	CO5	**Assessment:**	Assessment Tool 1 – **Research Paper Review / Literature Survey**
Weightage:	20%	**Total Marks:**	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
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Code of Conduct

I Tarunika S (192511024) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Tarunika S

1. Introduction

High-speed rail (HSR) networks are rapidly expanding worldwide, and operators are under increasing pressure to move beyond safe and punctual transport toward a connected, service-oriented travel experience. A Smart High-Speed Rail Passenger Experience Platform aims to integrate real-time information, onboard connectivity, personalization and IoT-based sensing to improve comfort, convenience and satisfaction for passengers before, during and after their journey. This survey reviews recent research on IoT, AI and communication technologies applied to railways, with the specific objective of understanding how existing work addresses — or fails to address — the passenger-experience dimension of smart rail systems, and where opportunities remain for a dedicated experience-platform design.

2. Literature Review

2.1 Self-Powered Sensing and AIoT in Rail Transit

A 2024 review examined self-powered and self-sensing devices used across rail transit scenarios, situating them within the broader convergence of AI and IoT referred to as AIoT. The study surveyed piezoelectric and triboelectric sensing technologies and their role in supporting energy-efficient, autonomous train operation. While the review is thorough on infrastructure health-monitoring, it treats passenger comfort only indirectly, through vibration and environmental sensing, without proposing a passenger-facing service layer.

2.2 6G-Enabled Smart Railways

A 2025 survey explored how sixth-generation (6G) wireless networks could support smart railway operations, covering integrated network architectures, sensing-communication convergence and edge computing. The authors argue that 6G is essential to deliver a fully connected, real-time service experience for passengers on high-speed trains, referencing European and Chinese railway modernization roadmaps. The work remains largely architectural and forward-looking, offering no concrete application design for passenger-facing services that such connectivity would enable.

2.3 IoT System Architecture for High-Speed Railways

Earlier work proposed a layered IoT system architecture for high-speed rail, covering device, network, gateway and platform layers, and discussed how this architecture could extend IoT applications across multiple areas of the HSR industry, including safety and infrastructure monitoring. Related work by Jo and Kim proposed a similarly structured, cost-effective IoT device-and-platform solution for smart railway infrastructure. These architectures provide a technically sound foundation but were designed primarily around operational and maintenance functions rather than passenger interaction.

2.4 AI, IoT and Digital Twin Convergence in Railways

A recent systematic review evaluated 199 published studies on the integration of AI, IoT, digital twin technology and autonomous systems in railways. It found that this convergence supports predictive maintenance, energy optimization and enhanced passenger experience, while identifying persistent challenges in cybersecurity, regulatory compliance and legacy-infrastructure integration. Although passenger experience is acknowledged as a benefit, the review notes it is rarely the primary focus of the studies surveyed, most of which center on operational or safety outcomes.

2.5 RFID-Based Passenger Data Analytics

A 2023 study used RFID tags with Arduino-based hardware to collect passenger travel data on an IoT-enabled railway system, classifying passengers into regular, irregular and habitual travel categories based on temporal and spatial patterns. This clustering approach demonstrates the value of low-cost sensing for building a picture of passenger behaviour, which could inform personalized services, but the prototype was tested at a small scale and was not connected to any live passenger-facing application.

3. Comparative Analysis

The table below compares the reviewed studies across research problem, methodology, tools/datasets used, key findings, limitations and the specific research gap relevant to a passenger-experience platform.

Author / Year	Research Problem	Methodology	Dataset / Tools	Key Findings	Limitations	Research Gap
Zhang et al., 2024	Lack of energy-autonomous sensing for continuous rail condition and passenger-environment monitoring	Review of self-powered sensing devices integrated with AIoT (AI + IoT) frameworks	Piezoelectric / triboelectric sensor prototypes; ML/DL algorithms	Self-powered sensors reduce maintenance dependency and enable real-time onboard data collection	Mostly infrastructure-focused; limited discussion of passenger-facing applications	No unified platform linking sensing data to passenger experience services
Wang et al., 2025 (6G-Enabled Smart Railways)	Existing 4G/5G networks cannot meet ultra-reliable, low-latency needs of high-speed rail services	Survey of 6G architecture, ISAC, edge computing and network slicing for railways	Simulation-based architecture proposals; literature synthesis	6G can enable a fully connected, real-time passenger service experience onboard high-speed trains	Technology is largely conceptual; commercial 6G deployment is still years away	Absence of a concrete passenger-experience application layer built on next-gen connectivity
Jo & Kim, 2021 (HSR IoT Architecture)	Fragmented IoT adoption across high-speed rail subsystems	Proposed layered IoT system architecture (device, network, platform layers)	IoT gateway and platform server prototype	Unified architecture improves scalability of railway IoT deployment	Focused on operational/safety IoT, not passenger comfort or personalization	Passenger-experience layer not addressed within the architecture
Recent Review, 2024–25 (AI/IoT/Digital Twin in Rail)	Fragmented adoption of AI, IoT and digital twins across 199 reviewed studies	Systematic literature review across thematic domains	199 published studies (secondary data)	AI-IoT-digital twin convergence improves predictive maintenance and passenger experience	Cybersecurity, legacy-system integration and regulatory gaps remain unresolved	Few studies evaluate passenger-experience metrics as a standalone research theme
Case Study, Swedish Railways (Mobile Real-Time Info)	Passenger discomfort from disruptions due to lack of timely travel information	Evaluation model quantifying value of real-time information under disruption scenarios	Historical delay distribution data, commuter-line case study	Real-time pre-trip and on-trip information significantly reduces perceived wait-time cost	Model validated only for one regional commuter line; not high-speed rail specific	Limited work on multimodal, personalized real-time notification platforms for HSR passengers

4. Identified Research Gaps

- Most IoT and AI railway architectures are optimized for operations, maintenance and safety — a dedicated passenger-experience application layer is largely absent.
- Passenger behaviour analytics (e.g., RFID-based clustering) exist in isolated prototypes but are not connected to real-time, personalized service delivery.
- Next-generation connectivity research (5G/6G) discusses passenger-experience potential conceptually, without a working platform design or evaluation.
- Very few studies combine real-time disruption information, onboard connectivity and personalization into a single, high-speed-rail-specific system, and fewer still evaluate passenger satisfaction as a measurable outcome.

5. Future Research Directions

- Design a unified passenger-experience platform that fuses IoT sensor data, AI-based personalization and real-time communication (5G/6G) specifically for high-speed rail.
- Extend RFID/behavioural-clustering approaches into live recommendation engines for seat comfort, catering, and journey information tailored to passenger type.
- Develop lightweight edge-computing solutions so real-time disruption alerts and personalized content can be delivered with low latency at high train speeds.
- Establish standardized metrics for evaluating passenger satisfaction and experience quality, enabling fair comparison across future smart-rail platforms.

6. Conclusion

The reviewed literature shows steady progress in applying IoT, AI, self-powered sensing and advanced wireless technologies to high-speed rail, primarily to improve safety, maintenance and network performance. Passenger experience is frequently mentioned as a motivating outcome but is seldom the direct subject of system design or evaluation. This gap indicates a clear opportunity for a Smart High-Speed Rail Passenger Experience Platform that integrates existing sensing, connectivity and analytics research into a passenger-centric system, supported by measurable experience metrics.